

Logistic Regression

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1 Introduction

Logistic Regression is one of the most fundamental supervised machine learning algorithms used for **classification**. Unlike Linear Regression, which predicts continuous numerical values, Logistic Regression predicts the **probability** that an observation belongs to a particular class.

Although its name contains the word *regression*, Logistic Regression is primarily a classification algorithm. The term “regression” comes from the fact that the algorithm first computes a linear regression model before transforming its output into a probability using the sigmoid function.

Logistic Regression is mainly used for binary classification problems where the target variable has only two possible classes.

Typical examples include

- Spam or Not Spam
- Healthy or Diseased
- Fraud or Legitimate
- Pass or Fail
- Approved or Rejected

2 Binary Classification

Binary classification consists of assigning each observation to one of two possible classes.

Usually, these classes are represented numerically as

$$0 \text{ and } 1$$

where

- Class 0 represents the negative class.
- Class 1 represents the positive class.

For example,

The objective of Logistic Regression is to estimate the probability

$$P(Y = 1|X),$$

which represents the probability that an observation belongs to the positive class given the input features.

3 Why Logistic Regression?

Suppose we want to predict whether a customer will purchase a product.

The dataset contains

The desired output is not a numerical value.

Instead, it belongs to one of two categories:

Purchased

or

NotPurchased.

Therefore, the prediction should represent the probability that a customer belongs to the positive class.

For example,

Notice that probabilities must always satisfy

$$0 \leq P \leq 1.$$

This requirement motivates the use of Logistic Regression.

4 Why Not Linear Regression?

One may wonder why Linear Regression cannot simply be used for classification.

Recall that Linear Regression predicts

$$y = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n.$$

The output of this equation may take any real value.

For example,

$$-5, -1, 0, 2, 15, 100.$$

There is no restriction on the predicted value.

This behavior is perfectly suitable for problems such as

- House price prediction

- Temperature prediction
- Population estimation
- Stock price prediction

because these quantities are continuous.
 However, classification problems require probabilities.
 A probability must satisfy

$$0 \leq P \leq 1.$$

Linear Regression cannot guarantee this property.

4.1 Example

Suppose a Linear Regression model predicts

$$\hat{y} = 1.34.$$

Does this represent
 134% probability?
 Obviously not.
 Now suppose another prediction is

$$\hat{y} = -0.25.$$

Does this represent
 -25% probability?
 Again, the answer is no.

Therefore, Linear Regression cannot directly produce meaningful probabilities for classification tasks.

4.2 The Need for a Probability Function

Instead of using the linear model directly,
 Logistic Regression computes

$$z = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n.$$

Notice that the variable is now denoted by

$$z,$$

not

$$y.$$

This value is called the **linear score**.

The linear score can take any real value,

$$-\infty < z < +\infty.$$

Instead of interpreting this score directly,

Logistic Regression transforms it into a probability using a special mathematical function called the **Sigmoid Function**.

The prediction process becomes

$$Features \rightarrow LinearScore \rightarrow SigmoidFunction \rightarrow Probability$$

The sigmoid function guarantees that every output lies between 0 and 1, making it suitable for probability estimation.

5 Transition to the Sigmoid Function

We have established that Linear Regression cannot be used directly for classification because its predictions are not bounded between 0 and 1.

The natural question is therefore:

How can we transform any real number into a valid probability?

The answer is provided by the **Sigmoid Function**, which forms the core of Logistic Regression and will be studied in the next section.

6 The Sigmoid Function

The sigmoid function is the core component of Logistic Regression. Its primary purpose is to transform the linear score into a probability that lies between 0 and 1.

Mathematically, the sigmoid function is defined as

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where

- z is the linear score,
- $e \approx 2.71828$ is Euler's number,
- $\sigma(z)$ is the estimated probability.

Unlike the linear model, whose output can be any real number, the sigmoid function always produces

$$0 < \sigma(z) < 1.$$

Consequently, every prediction can be interpreted as a probability.

6.1 Why Do We Need the Sigmoid Function?

The linear model produces

$$z = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n.$$

The value of z may be

$$-100, \quad -5, \quad 0, \quad 4, \quad 500.$$

These values cannot be interpreted as probabilities.

The sigmoid function converts every possible value of z into a valid probability.

Conceptually,

$$z \longrightarrow \sigma(z)$$

where

$$-\infty < z < +\infty$$

becomes

$$0 < \sigma(z) < 1.$$

6.2 Numerical Examples

Example 1 Suppose

$$z = 0.$$

Then

$$\sigma(0) = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5.$$

The model is completely uncertain and assigns equal probability to both classes.

Example 2 Suppose

$$z = 2.$$

Then

$$e^{-2} \approx 0.1353.$$

Therefore,

$$\sigma(2) = \frac{1}{1 + 0.1353} = 0.8808.$$

Interpretation:

There is an 88.08% probability that the observation belongs to the positive class.

Example 3 Suppose

$$z = -2.$$

Then

$$e^{-(-2)} = e^2 \approx 7.389.$$

Hence,

$$\sigma(-2) = \frac{1}{1 + 7.389} = 0.1192.$$

Interpretation:

The probability of belonging to the positive class is only 11.92%.

6.3 Important Properties

The sigmoid function has several important mathematical properties.

- Its output is always between 0 and 1.
- It is continuous.
- It is differentiable.
- It is monotonically increasing.
- Large positive values approach 1.
- Large negative values approach 0.
-

$$\sigma(0) = 0.5.$$

6.4 Common Sigmoid Values

Observe that increasing z also increases the predicted probability.

7 The Linear Model

Before applying the sigmoid function, Logistic Regression first computes a linear combination of the input features.

This quantity is called the **linear score** and is denoted by z .

It is computed as

$$z = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n.$$

In matrix notation,

$$z = \mathbf{w}^T \mathbf{x} + b.$$

where

Table 1: Examples of binary classification problems

Application	Classes
Email Filtering	Spam / Not Spam
Medical Diagnosis	Healthy / Diseased
Loan Approval	Approved / Rejected
Fraud Detection	Legitimate / Fraud
Education	Pass / Fail

Table 2: Example dataset

Age	Salary	Purchased
22	30000	No
30	50000	Yes
45	90000	Yes
20	18000	No

Table 3: Probability interpretation

Probability	Prediction
0.95	Purchased
0.82	Purchased
0.40	Not Purchased
0.10	Not Purchased

Table 4: Typical sigmoid outputs

z	$\sigma(z)$
-5	0.0067
-2	0.1192
-1	0.2689
0	0.5000
1	0.7311
2	0.8808
5	0.9933

- $\mathbf{x}$ is the feature vector,
- $\mathbf{w}$ is the vector of model weights,
- b is the bias,
- z is the linear score.

Notice that this equation is identical to the one used in Linear Regression. The difference is that Logistic Regression does not stop here. Instead,

$$z \longrightarrow \sigma(z) \longrightarrow P(Y = 1|X).$$

7.1 Input Features

Features describe the characteristics of each observation.

For example, when predicting whether a patient has heart disease, the features may include

- Age
- Blood Pressure
- Cholesterol
- Body Mass Index (BMI)

Each patient therefore has a feature vector

$$\mathbf{x} = (x_1, x_2, x_3, x_4).$$

7.2 Weights

Each feature contributes differently to the prediction.

The coefficients

$$w_1, w_2, \dots, w_n$$

are called the model weights.

A large positive weight increases the probability of the positive class.

A negative weight decreases that probability.

Therefore, the weights determine how influential each feature is during prediction.

7.3 Bias

The bias, denoted by b , represents the intercept of the model.

It shifts the decision boundary without depending on any input feature.

Even if all feature values are zero, the bias still contributes to the linear score.

8 Worked Example

Suppose a Logistic Regression model has learned

$$w_1 = 0.04,$$

$$w_2 = 0.00008,$$

and

$$b = -5.$$

Assume a customer has

- Age = 30
- Salary = 60000

Step 1: Compute the Linear Score

Age contribution

$$0.04 \times 30 = 1.2.$$

Salary contribution

$$0.00008 \times 60000 = 4.8.$$

Adding the bias,

$$z = 1.2 + 4.8 - 5 = 1.$$

Therefore,

$$z = 1.$$

Step 2: Apply the Sigmoid Function

$$\sigma(1) = \frac{1}{1 + e^{-1}} = 0.7311.$$

Therefore,

$$P(Y = 1|X) = 0.7311.$$

The model estimates a probability of

$$73.11\%.$$

Step 3: Final Prediction

Suppose the decision threshold is

$$0.5.$$

Since

$$0.7311 > 0.5,$$

the observation is classified as belonging to the positive class.

9 Prediction Pipeline

The complete prediction process of Logistic Regression can be summarized as

InputFeatures $\rightarrow$ *LinearModel* $\rightarrow$ *LinearScore*(z) $\rightarrow$ *SigmoidFunction* $\rightarrow$ *Probability* $\rightarrow$ *DecisionThreshold*

Every prediction made by Logistic Regression follows this sequence of operations.

10 Conclusion

At this stage, we understand how Logistic Regression transforms input features into probabilities. However, one important question remains unanswered:

How does the algorithm determine the optimal values of the weights and the bias?

The answer lies in the optimization process, where Logistic Regression minimizes a cost function using Gradient Descent. These concepts will be presented in the following sections.